

Tax Evasion and Productivity

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July 27, 2026

Recap

Last meeting: What is an interesting question to wrap up the JMP?

- ▶ Counterfactual question: Optimal tax rate

Today:

- ▶ SOC of firm's optimization problem
- ▶ Functional forms for detection probability and tax evasion cost
- ▶ Estimation strategy

Introduction

- ▶ What's the question?
- ▶ Why should you care?

Contribution

- ▶ What do I do?
- ▶ How do I do it?

Overview of Results

- ▶ What do I find?
- ▶ Why is it important?

Colombia

Model

The Model

$$\begin{aligned} \max_{K_{it}, L_{it}, M_{it}, e_{it}} \quad & (1 - q(e_{it})) \left[P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t M_{it} - r_t K_{it} - w_t L_{it} \right. \\ & \left. - \tau (P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t (M_{it} + e_{it})) \right] \\ & + q(e_{it}) \left[(1 - \tau) (P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t M_{it}) - r_t K_{it} - w_t L_{it} \right. \\ & \left. - \phi \rho_t \tau e_{it} \right] - \kappa_{it}(e_{it}, \omega_{it}) \\ \text{s.t.} \quad & Y_{it} = K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^\beta \exp(\omega_{it} + \varepsilon_{it}) \end{aligned} \tag{1}$$

Important Assumptions

- ▶ Following seminal work of Allingham and Sandmo (1972), tax authority audits firms at random and always detects tax evasion
 - ▶ The probability of detection, $q(e_{it})$, is a function of the level of tax evasion, e_{it} . It might depend on other exogenous variables such as the monitoring technology of the tax authority, for example.
 - ▶ Impose assumption from Empirical model which assumes that choosing M does not enter the probability of detection or the cost of evasion. In other words, choosing M is independent of choosing e.

Important Assumptions (Cont.)

The cost of evasion, $\kappa_{it}(e_{it}, \omega_{it})$: any resources of the firm allocated to the tax evasion activity that could have been employed in a productive activity (e.g. manager's time spent on tax evasion instead of managing the firm).

- ▶ Unlike the penalty of tax evasion, ϕ , it does not depend on whether you are detected or not
- ▶ Increasing in the level of tax evasion, e_{it}
- ▶ Decreasing in firm's productivity, ω_{it} , but spikes at the top of the productivity distribution.

Important Assumptions - discussion on κ

- ▶ More productive firms have more resources to allocate to tax evasion, but since huge firms have access to more sophisticated legal tax avoidance strategies, the cost of illegal tax evasion is higher for the most productive firms.
- ▶ It could also be the case that the probability of detection is flat with respect to productivity except for huge (top productive) firms which could have a probability close to 1.
- ▶ Empirically, without data on cost of evasion or auditing information, q and κ are not separately identified. During estimation, we assume functional forms that allow us to reproduce the empirical patterns of tax evasion across the productivity distribution.

Model (Cont.)

The firms' problem (Equation 1) can be rewritten as follows,

$$\begin{aligned} \max_{K_{it}, L_{it}, M_{it}, e_{it}} \quad & \mathbb{E}[\pi_{it} | \Gamma] = (1 - \tau) (P_t \mathbb{E}[Y_{it}] - \rho_t M_{it}) - r_t K_{it} - w_t L_{it} \\ & + [1 - q(e_{it})] (\rho_t \tau e_{it}) - q(e_{it}) \phi \rho_t \tau e_{it} - \kappa_{it}(e_{it}, \omega_{it}) \\ \text{s.t.} \quad & Y_{it} = K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta} \exp(\omega_{it} + \varepsilon_{it}) \end{aligned} \tag{2}$$

First Order Condition (FOC)

An interior solution leads to following first-order conditions,

$$\beta K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{-(1-\beta)} \exp(\omega_{it}) = \frac{\rho_t}{P_t} \quad (3)$$

$$\tau \rho_t (1 - (1 + \phi) (q(e_{it}) + q'(e_{it})e_{it})) = \frac{\partial \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}} \quad (4)$$

(ε_{ijt} is measurement error, so $\mathcal{E} \equiv E[\exp(\varepsilon_{ijt})] = 1$)

Tax Evasion

To better understand the incentives to overreport inputs, we can rewrite Equation 4 in terms of the evasion elasticity of the probability of detection, $\xi_{q,e} = \frac{\partial q(e_{it})}{\partial e_{it}} \frac{e_{it}}{q(e_{it})}$, as follows,

$$\tau \rho_t \left(1 - (1 + \phi) (1 + \xi_{q,e}) q(e_{it}) \right) = \frac{\partial \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}} \quad (5)$$

Corner Solution

Note that some firms will optimally choose not to evade, $e_{it} = 0$.

For these firms, the marginal cost of tax evasion is greater than the marginal benefit for any non-zero level of tax evasion.

This implies that for these firms, Equation 4 becomes,

$$\tau \rho_t \leq \frac{\partial \kappa_{it}(0, \omega_{it})}{\partial e_{it}} \quad (6)$$

Second Order Condition (SOC)

To guarantee a unique solution, the hessian matrix of the firm's problem must be negative definite.

$$\nabla^2 \mathbb{E}[\pi_{it} | \Gamma] = \begin{bmatrix} \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial e_{it}^2} & 0 & 0 & 0 \\ 0 & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial M_{it}^2} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial M_{it} \partial L_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial M_{it} \partial K_{it}} \\ 0 & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial L_{it} \partial M_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial L_{it}^2} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial L_{it} \partial K_{it}} \\ 0 & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial K_{it} \partial M_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial K_{it} \partial L_{it}} & \frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial K_{it}^2} \end{bmatrix} \quad (7)$$

- ▶ For a negative definite hessian matrix, the leading principal minors must alternate signs,
 - ▶ odd order minors must be negative and even-order minors positive.
- ▶ In our case, since
 - ▶ 1) the production function is strictly concave (CD) and
 - ▶ 2) choosing the level of tax evasion does not depend on the production function inputs; second cross derivatives are zero,
 - ▶ we would only have to check the first leading principal minor —the second derivative with respect to e_{it} — is negative,

$$\frac{\partial^2 \mathbb{E}[\pi_{it} | \Gamma]}{\partial e_{it}^2} = -\tau \rho_t (1 + \phi) (2q'(e_{it}) + e_{it} q''(e_{it})) - \frac{\partial^2 \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}^2} < 0 \quad (8)$$

For strict concavity, we need only one of the following two conditions to be strictly positive,

$$\frac{\partial^2 \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}^2} \geq 0 \quad (9)$$

$$\tau \rho_t (1 + \phi) (2q'(e_{it}) + e_{it} q''(e_{it})) \geq 0 \quad (10)$$

The second condition can be simplified to

$$-\frac{q''(e_{it})}{q'(e_{it})}e_{it} < 2 \quad (11)$$

since economic theory implies that $q'(e_{it}) > 0$ and $e_{it} > 0$, this condition imposes a **restriction on the curvature** of the detection function, $q(e_{it})$

- ▶ **Linear** detection **always** satisfies the SOC, since $q''(e_{it}) = 0$.
- ▶ **Convex** also satisfies the SOC **always**, since $q''(e_{it}) > 0$.
- ▶ **Concave** works if not too concave, i.e. **inequality holds**.

What does this inequality imply in practice?

With $\phi = 0$, logs of the interior FOC $\tau\rho_t(1 - q - q'e) = \partial\kappa/\partial e$ define

- ▶ the **log marginal benefit** $h(e) \equiv \ln(\tau\rho_t) + \ln(1 - q(e) - q'(e)e)$,
- ▶ with FOC $h(e) = \ln(\partial\kappa/\partial e)$.

	$q(e)$	SOC	$h(e) - \ln(\tau\rho_t)$	FOC ceiling
Linear	$\min\{\lambda e, 1\}$	$0 < 2$	$\ln(1 - 2\lambda e)$	$\lambda e < \frac{1}{2}$
Concave	$1 - e^{-\lambda e}$	$\lambda e < 2$	$-\lambda e + \ln(1 - \lambda e)$	$\lambda e < 1$
Convex	$\min\{(\lambda e)^k, 1\}$ $k > 1$	$1 - k < 2$	$\ln(1 - (1+k)(\lambda e)^k)$	$\lambda e < (1+k)^{-1/k}$

- ▶ All satisfy the SOC on their interior (exp's $\lambda e < 2 \iff \lambda e < 1$);
- ▶ All impose a restriction on λe .

A detection technology: the audit threshold

- ▶ The firm chooses overreporting e .
- ▶ The tax authority draws a threshold $T \sim U[0, 1/\lambda]$ and audits the firm if $e \geq T$.
 - ▶ An audit always uncovers evasion (Allingham and Sandmo 1972).
 - ▶ T is random to the firm (audit selection, third-party cross-checks, which inspector).

$$q(e) = \Pr(T \leq e) = \min\{\lambda e, 1\}.$$

- ▶ **Threshold scale:** $1/\lambda$ is the largest overreporting any audit draw tolerates; the *average* audit-triggering threshold is $1/2\lambda$.
- ▶ **Evasion bound:** the FOC stops firms at $\lambda e < \frac{1}{2}$
 - ▶ the most aggressive evader overreports below the *average* audit-triggering threshold,
 - ▶ at threshold, $q(1/2\lambda) = 50\%$ (a coin flip)

If $q(e)$ is concave, $T \sim \text{Exp}(\lambda)$ and $q(e) = 1 - e^{-\lambda e}$, mean threshold $1/\lambda$, detection $\approx 63\%$ at $e = 1/\lambda$.

How likely is the most aggressive evader caught?

At the FOC ceiling, the largest evasion drives the marginal benefit to zero,
 $q(b) + q'(b)b = 1$.

	b	$q(b)$	$\xi_{q,e} _{e=b}$	$E[T]$
Linear	$\frac{1}{2\lambda}$	50%	1 (unit elastic)	$\frac{1}{2\lambda}$
Exponential	$\frac{1}{\lambda}$	63%	$\frac{\lambda b}{e^{\lambda b} - 1} < 1$ (inelastic)	$\frac{1}{\lambda}$
Power	$\frac{1}{\lambda(1+k)^{1/k}}$	$\frac{1}{1+k} < 50\%$	$k > 1$ (elastic)	$\frac{k}{\lambda(k+1)}$

- ▶ $\xi_{q,e} = q'(e)e/q(e)$ is the evasion elasticity of the detection probability
- ▶ $\frac{1}{e-1} = 0.58$

Discussion

- ▶ Power (convex) form produces the lowest average audit-triggering threshold,
 - ▶ Exponential (concave) the largest.
 - ▶ Linear is intuitive: at max evasion, the tax authority flips a coin to audit/caught the firm
- ▶ An increase in 1% in the level of evasion increases the probability of being caught, by
 - ▶ 1% if linear (constant)
 - ▶ more than 1% if convex (constant)
 - ▶ less than 1% if concave (increasing), $\xi_{q,e}|_{e=0} = 0$ and $\xi_{q,e}|_{e=1/\lambda} = 0.58$.
- ▶ The extra dollar the firm hides increase its probability of detection more than the previous dollar.
- ▶ For linear and exponential, the FOC bounds evasion *at* the expected value of T
 - ▶ For power, FOC bounds evasion *below* its expected value: At $k = 2$, $\frac{1}{\lambda\sqrt{3}} < \frac{2}{\lambda 3}$

Estimation and recommendation

- ▶ **Linear** and **exponential** carry a single parameter λ
- ▶ **Power** $(\lambda e)^k$ adds a parameter, curvature k

Do **linear** and **power**,

- ▶ Start with **linear** ($k = 1$). Linear is the $k = 1$ case of the power form (ceiling $(1+k)^{-1/k} = \frac{1}{2}$).
- ▶ Then, estimate power (λ, k) , imposing feasibility $\lambda e < (1+k)^{-1/k}$ for *every* firm. Test if $k = 1$.

Empirical Strategies

- ▶ Testing for the presence of tax evasion by industry
 - ▶ Robust to different specifications
- ▶ Regression
 - ▶ Useful to study tax evasion heterogeneity across firms and over time
- ▶ Deconvolution
 - ▶ Recover distribution of tax evasion and productivity
 - ▶ Estimation of unbiased Production Function

Data

Testing

Empirical Strategy

Functional Forms

The probability of detection, $q(e_{it})$, and the cost of evasion, $\kappa_{it}(e_{it}, \omega_{it})$, functions should display the following properties to be consistent with the observed patterns in the data and with theory:

- ▶ $q(e_{it}) = \min\{\lambda e, 1\}$
 - ▶ $q(e_{it}) \in [0, 1]$, $q(0) = 0$, $q(\infty) = 1$, and $q(e_{it})$ is increasing in e . Since I do not have variation in government audit effort, λ is treated as constant across firms and time
- ▶ $\kappa_{it}(e_{it}, \omega_{it}) = e_{it} \mathbf{e}^{\delta_0 - \delta_1 \omega_{it} + \delta_2 \omega_{it}^2 + \psi_{it}}$
 - ▶ $\kappa_{it}(e_{it}, \omega_{it}) \in [0, \infty)$, $\kappa_{it}(0, \omega_{it}) = 0$, $\frac{\partial \kappa_{it}(0, \omega_{it})}{\partial e_{it}} \geq 0$.
 - ▶ $\frac{\partial \kappa_{it}(e_{it}, \omega_{it})}{\partial \omega_{it}} \leq 0$ and $\frac{\partial^2 \kappa_{it}(e_{it}, \omega_{it})}{\partial \omega_{it}^2} \geq 0$ to be consistent with the observed patterns in the data.
 - ▶ U-shape not ideal, but simple and captures the idea of decreasing cost at high ω
 - ▶ Corporations out of the FOC
 - ▶ ψ_{it} idiosyncratic shock to the cost of evasion to explain tax evasion heterogeneity observed in empirical analysis

FOC

Using the functional forms defined above, the FOCs wrt tax evasion become for unincorporated firms,

$$\tau\rho_t(1 - 2\lambda e_{it}) = \mathbf{e}^{\delta_0 - \delta_1\omega_{it} + \delta_2\omega_{it}^2 + \psi_{it}}$$

Taking logs and defining

$$h(e_{it}) \equiv \ln(\tau\rho_t) + \ln(1 - \lambda 2e_{it}), \quad e_{it} \in (0, \frac{1}{2\lambda}),$$

the evasion FOC become

$$\psi_{it} = h(e_{it}) - \delta_0 + \delta_1\omega_{it} - \delta_2\omega_{it}^2 \tag{12}$$

How to estimate?

- ▶ $\tau\rho_t$ is *computed from data*

I do not observe e_{it} nor ω_{it} at the firm level.

- ▶ However, using the estimated **parameters of the production** function, we can
- ▶ rewrite the FOC in terms of data, parameters, and
- ▶ *only one* independent unobservable ε

How to estimate the production function parameters and the distribution of ε ? **Same as before**

Estimating β

Same as before: From Corporations, the truth-reporters, I recover

- ▶ the output elasticity of materials, β , and the
- ▶ distribution of the error term, f_ε ,

For Corporations, $M_{it}^* = M_{it}$, and thus,

$$\ln \left(\frac{\rho_t M_{it}^*}{P_t Y_{it}} \right) = \ln(\beta) - \varepsilon_{it}$$

Production Function Estimation

- Having recovered β , for everyone, I can form the following variable,

$$\begin{aligned}\mathcal{V}_{it} &\equiv \ln \left(\frac{\rho_t M_{it}^*}{P_t Y_{it}} \right) - \ln(\beta) \\ &= \ln \left(\frac{\rho_t M_{it}^*}{P_t Y_{it}} \right) - \ln \left(\frac{\rho_t M_{it}}{P_t Y_{it}} \right) + \ln \left(\frac{\rho_t M_{it}}{P_t Y_{it}} \right) - \ln(\beta) \\ &= \ln \left(\frac{M_{it}^*}{M_{it}} \right) - \varepsilon_{it}\end{aligned}\tag{13}$$

Production Function Estimation (Cont.)

- To recover the remaining production function parameters, the **original strategy is still valid**

To see why, note that,

$$\begin{aligned}\mathcal{W}_{it} &\equiv y_{it} - \beta(m_{it}^* - \mathcal{V}_{it}) \\ &= y_{it} - \beta(m_{it}^* - \ln\left(\frac{M_{it}^*}{M_{it}}\right) + \varepsilon_{it}) \\ &= y_{it} - \beta(m_{it}^* - \ln(M_{it}^*) + \ln(M_{it})) - \beta\varepsilon_{it} \\ &= y_{it} - \beta m_{it}^* + \beta \ln(M_{it}^*) - \beta \ln(M_{it}) - \beta\varepsilon_{it} \\ &= \alpha_K k_{it} + \alpha_L l_{it} + \omega_{it} + (1 - \beta)\varepsilon_{it}\end{aligned}\tag{14}$$

the last equality is *exactly the same* as the original discussed strategy

Using productivity's Markov process to recover remaining PF parameters

Therefore, the **previous approach is still valid** to recover the remaining production function parameters, productivity's Markov process, and the distribution of the productivity shock $(\alpha_K, \alpha_L, \omega_{it})$.

Assuming, productivity follows an AR(1) Markov process,

$$\omega_{it} = \gamma_0 + \gamma_1 \omega_{it-1} + \eta_{it} \quad (15)$$

we recover the PF parameters, by substituting Equation 14, into Equation 15,

$$\begin{aligned} \mathcal{W}_{it} = & \gamma_0 + \alpha_K k_{it} + \alpha_L l_{it} \\ & + \gamma_1 (\mathcal{W}_{it-1} - \alpha_K k_{it-1} - \alpha_L l_{it-1}) \\ & + (1 - \beta) \varepsilon_{it} - \gamma_1 (1 - \beta) \varepsilon_{it-1} + \eta_{it} \end{aligned} \quad (16)$$

Estimating Equation 16

By finding instruments, we can estimate Equation 16, using the conditional moment,

$$E[(1 - \beta)\varepsilon_{it} - \gamma_1(1 - \beta)\varepsilon_{it-1} + \eta_{it} | Z_{it}] = 0$$

- ▶ I use capital and labour as instruments for themselves and reported materials and $\mathcal{W}_{it-2}$ as instruments to control for endogeneity between $\mathcal{W}_{it-1}$ and ε_{it-1}
 - ▶ $Z = \{k_{it}, l_{it}, m_{it-1}^*, \mathcal{W}_{it-2}\}$

Using the estimated parameters of the production function

As previously discussed, with α_K and α_L at hand form,

$$\tilde{\mathcal{W}}_{it} \equiv \mathcal{W}_{it} - \alpha_K k_{it} - \alpha_L l_{it} = \omega_{it} + (1 - \beta)\varepsilon_{it} \quad (17)$$

and recover f_ω using deconvolution,

$$f_{\tilde{\mathcal{W}}} = \int f_\varepsilon \left(\frac{1}{1 - \beta} (\tilde{\mathcal{W}} - \omega) \right) f_\omega(\omega) d\omega$$

Solving for M and e

NEW From Equation 13, I can *algebraically solve* for true inputs as function of observed variables and an the error term,

► $M_{it} = M_{it}^* \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon_{it}}.$

Therefore,

$$\begin{aligned} e_{it} &= M_{it}^* - M_{it} \\ &= M_{it}^* (1 - \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon_{it}}) \end{aligned}$$

- ε_{it} is not observed firm-by-firm, only its distribution f_{ε}
- not a literal point recovery of e_{it} , but the map that could be evaluated for some draw of $\varepsilon^{(s)} \sim \hat{f}_{\varepsilon}$

Recast in terms of one unobservable

I can do the same and solve for ω_{it} from Equation 17

For each value of ε everything else is pinned down:

$$e_{it}(\varepsilon) = M_{it}^* (1 - \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon})$$

$$\omega_{it}(\varepsilon) = \tilde{\mathcal{W}}_{it} - (1 - \beta)\varepsilon$$

$$\psi_{it}(\varepsilon) = h(e_{it}(\varepsilon)) - \delta_0 + \delta_1 \omega_{it}(\varepsilon) - \delta_2 \omega_{it}(\varepsilon)^2.$$

► Estimate f_ψ and $\theta = \{\lambda, \delta_0, \delta_1, \delta_2\}$ by Maximum Simulated Likelihood (MSL)

- ▶ We need the density of what we actually **observe** $f_{\mathcal{V}, \tilde{\mathcal{W}}}(\mathcal{V}, \tilde{\mathcal{W}})$
- ▶ After the recast, one unobservable remains: ε_{it} , with known distribution f_{ε} (from stage 1)
- ▶ I would need a *density transformation* to get from the primitives' density $f_{\psi, \omega, \varepsilon}(\psi, \omega, \varepsilon)$ to the density of the observables,
 - ▶ accounting for the nonlinear map from ψ to $\mathcal{V}$ (Theorem 7.5.3, next slide) *and*
 - ▶ integrating out ε

$$f_{\mathcal{V}, \tilde{\mathcal{W}}}(\mathcal{V}, \tilde{\mathcal{W}}) = \int f_{\psi}(\psi(\varepsilon)) f_{\omega}(\omega(\varepsilon)) f_{\varepsilon}(\varepsilon) |\det J\Phi^{-1}(\varepsilon)| d\varepsilon$$

The object to estimate is f_{ψ} and $\theta = (\lambda, \delta_0, \delta_1, \delta_2)$, since f_{ω} and f_{ε} are already known

Theorem 7.5.3 Change of Variables in R^n Judd (1998)

Assume that $\Phi : X \subset R^n \rightarrow R^n$ is C^1 and that there is a C^1 inverse map $\Phi^{-1} : Y \subset R^n \rightarrow X$ where $Y = \Phi(X)$ and Y and X are open. If f is integrable on X , then

$$\int_Y f(y)dy = \int_X f(\Phi(x)) |\det J(\Phi)(x)| dx,$$

where $J(\Phi)(x)$ is the Jacobian of Φ at x .

Corollary: Density transformation

Let $f_X(x)$ be the joint pdf of X , B any region in the support of Y , and $A = \Phi^{-1}(B)$

For $Y = \Phi(X)$: $P(X \in A) = P(\Phi(X) \in \Phi(A)) = P(Y \in B)$. Apply the theorem with $\Psi \equiv \Phi^{-1}$ in place of Φ (Φ^{-1} is C^1 with C^1 inverse Φ) to $g = f_X \cdot \mathbf{1}_A$:

$$P(X \in A) = \int_A f_X(x) dx = \int_B f_X(\Phi^{-1}(y)) |\det J\Phi^{-1}(y)| dy.$$

Since B is arbitrary, and using $J\Phi^{-1}(y) = [J\Phi(x)]^{-1}$ (inverse function theorem) at $x = \Phi^{-1}(y)$, then, the joint pdf of Y is:

$$f_Y(y) = \frac{f_X(x)}{|\det J\Phi(x)|} \Big|_{x=\Phi^{-1}(y)} \quad (18)$$

Setting up Φ for this model

Primitives $X = (\psi, \omega, \varepsilon)$, independent, density

$$f_{\psi, \omega, \varepsilon}(\psi, \omega, \varepsilon) = f_{\psi}(\psi) f_{\omega}(\omega) f_{\varepsilon}(\varepsilon).$$

- ▶ Invert the FOC $\psi = h(e) - \delta_0 + \delta_1 \omega - \delta_2 \omega^2$ for e holding ω fixed. Since $h'(e) < 0$ throughout the domain, this defines a C^1 inverse $e^*(\psi, \omega)$ (implicit function theorem)
- ▶ Convert to the log-ratio using $u(e) = \ln((M_{it} + e)/M_{it})$, with M_{it} held fixed as e varies ($M \perp e \mid \omega$)
- ▶ Define $G(\psi, \omega) \equiv u(e^*(\psi, \omega))$, u_{it} written as a function of the primitives

$\Phi : (\psi, \omega, \varepsilon) \mapsto (\mathcal{V}, \tilde{\mathcal{W}}, a)$, padding with the auxiliary $a \equiv \varepsilon$ so $\Phi : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ matches the theorem's dimension requirement:

$$\mathcal{V} = G(\psi, \omega) - \varepsilon, \quad \tilde{\mathcal{W}} = \omega + (1 - \beta)\varepsilon, \quad a = \varepsilon$$

The Jacobian matrix

$$J\Phi = \frac{\partial(\mathcal{V}, \tilde{\mathcal{W}}, a)}{\partial(\psi, \omega, \varepsilon)} = \begin{pmatrix} G_\psi & G_\omega & -1 \\ 0 & 1 & \beta \\ 0 & 0 & 1 \end{pmatrix}$$

where $G_\psi \equiv \partial G / \partial \psi$ and $G_\omega \equiv \partial G / \partial \omega$

Determinant

Expand along column 1 (only G_ψ is nonzero there):

$$\det J\Phi = G_\psi \cdot \det \begin{pmatrix} 1 & 1 & -\beta \\ 0 & 1 & 1 \end{pmatrix} = G_\psi \cdot 1 = G_\psi$$

- ▶ G_ω never enters because every cofactor multiplying it comes from a zero in column 1.
- ▶ Mechanically: $\tilde{\mathcal{W}} = \omega + (1 - \beta)\varepsilon$ has **unit coefficient on ω** , a flat translation, so it contributes no curvature.
- ▶ A Jacobian only appears where the map is genuinely curved: the $\psi \rightarrow \mathcal{V}$ link through G , not $\omega \rightarrow \tilde{\mathcal{W}}$.

Unpacking G_ψ

$G = u \circ e^*$, so by the chain rule $G_\psi = u'(e^*) \cdot e_\psi^*$.

► $u'(e)$: from $u(e) = \ln(M_{it} + e) - \ln M_{it}$, with M_{it} fixed,

$$u'(e) = \frac{1}{M_{it} + e}$$

evaluated at $e = e^*$, $M_{it} + e^* = M_{it}^*$, $\implies u'(e) = 1/M_{it}^*$

► e_ψ^* : differentiate the FOC identity $\psi = h(e^*(\psi, \omega)) - \delta_0 + \delta_1\omega - \delta_2\omega^2$ w.r.t. ψ (implicit function theorem, valid since $h' \neq 0$):

$$1 = h'(e^*) \cdot e_\psi^* \implies e_\psi^* = \frac{1}{h'(e^*)}$$

Combine:

$$G_\psi = \frac{1}{M_{it}^* h'(e)}$$

Assembling $J_i(\varepsilon)$

From Equation 18, the multiplying factor is $1/|\det J\Phi| = 1/|G_\psi|$:

$$J_i(\varepsilon) \equiv \frac{1}{|G_\psi|} = |h'(e)| M_{it}^*$$

Every piece is traceable: - $|h'(e)|$ from inverting the FOC ($\psi \rightarrow e$), - M_{it}^* from the log-ratio definition ($e \rightarrow \mathcal{V}$, holding the *true* M_{it} fixed) - they multiply because $G = u \circ e^*$ is a **composition**, so the chain rule stacks them.

Marginalizing out the auxiliary

Applying the density formula gives the joint density of $(\mathcal{V}, \tilde{\mathcal{W}}, a)$, but only $(\mathcal{V}, \tilde{\mathcal{W}})$ are observed;

- ▶ Integrating a out (an ordinary single integral, not part of the theorem) gives the likelihood contribution of firm i :

$$f_{\mathcal{V}, \tilde{\mathcal{W}}}(\mathcal{V}_{it}, \tilde{\mathcal{W}}_{it}) = \int f_{\psi}(\psi(\varepsilon)) J_i(\varepsilon) f_{\omega}(\omega(\varepsilon)) f_{\varepsilon}(\varepsilon) d\varepsilon \quad (19)$$

A 3-for-3 change of variables that ends in a 1-D integral:

- ▶ the theorem handles the square transformation and hands back a Jacobian;
- ▶ marginalizing the unobserved auxiliary coordinate is the separate, ordinary step left over once the theorem's job is done.

Simulated log-likelihood

$$\mathcal{L}(\theta, f_\psi) = \sum_{i \in \text{Unincorp}} \ln \left(\int f_\psi(\psi(\varepsilon)) J_i(\varepsilon) f_\omega(\omega(\varepsilon)) f_\varepsilon(\varepsilon) d\varepsilon \right) \quad (20)$$

Simulating the likelihood

Approximate Equation 19 by **Monte Carlo**

$$\widehat{\mathcal{L}}_i(\theta) = \frac{1}{S} \sum_{s=1}^S f_{\psi}(\psi(\varepsilon^{(s)})) J_i(\varepsilon^{(s)}) f_{\omega}(\omega(\varepsilon^{(s)})), \quad \varepsilon^{(s)} \overset{iid}{\sim} \hat{f}_{\varepsilon} \quad (21)$$

- ▶ Draws $\varepsilon^{(s)}$ come from *exactly* the density $\hat{f}_{\varepsilon}(\varepsilon) d\varepsilon$ multiplying the integrand — this is plain Monte Carlo integration, unbiased for **any** $S \geq 1$
- ▶ Alternatively, use **deterministic quadrature** to integrate Equation 19

Why this works despite ε entering e nonlinearly

- ▶ $e(\varepsilon) = M_{it}^* (1 - \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon})$ is nonlinear in ε
- ▶ But Equation 21 averages the **density value** $f_{\psi}(\psi(\varepsilon^{(s)})) J_i(\varepsilon^{(s)}) f_{\omega}(\omega(\varepsilon^{(s)}))$ at each draw
 - ▶ never averages $\psi(\varepsilon^{(s)})$ *by itself*
- ▶ Monte Carlo integration of a (possibly wild) function against its own sampling density is unbiased regardless of the function's shape
 - ▶ nonlinearity only breaks things if the reweighting is skipped
- ▶ Every simulated $\varepsilon^{(s)} \neq$ the firm's true ε_{it} , almost surely
 - ▶ but draws implying an implausible (ψ, ω) pair get down-weighted by $f_{\psi} \cdot J_i \cdot f_{\omega}$; only draws near the truth carry weight,
 - ▶ so the weighted average reconstructs the correct posterior-weighted integral even though no single draw is the truth

Convergence — a known theorem

Lee (1992): if the simulator of a likelihood contribution is (i) unbiased and continuous in θ , and (ii) built from draws i.i.d. across firms and replicates, then under standard M-estimation regularity (compactness, identification, dominance) the simulated-MLE is consistent and $\sqrt{N}$ -equivalent to the infeasible exact MLE, **provided** $\sqrt{N}/S \rightarrow 0$ as $N, S \rightarrow \infty$ (fixed S leaves an $O(1/S)$ bias, since $\ln(\text{average}) \neq \text{average of } \ln$)

- ▶ Unbiasedness (i): shown above, holds by construction
- ▶ Continuity (ii): h , q , and $J_i(\varepsilon)$ are smooth throughout the interior domain $e \in (0, \frac{1}{2\lambda})$
- ▶ Draws i.i.d. across i, t, s , independent of θ : by construction

An even cleaner argument: quadrature

- ▶ The preferred implementation is **deterministic quadrature**, not Monte Carlo simulation, for this 1-D integral
- ▶ Quadrature avoids simulation noise because it is deterministic and can be driven arbitrarily small by adding nodes
- ▶ The $\sqrt{N}/S \rightarrow 0$ condition above is then moot: with negligible quadrature error, ordinary MLE consistency applies directly

Caveat: this is a two-step estimator

- ▶ β and f_ε are **first-stage estimates**, not the truth
- ▶ Still consistent under standard conditions, but naive stage-2 standard errors ignore stage-1 estimation uncertainty
- ▶ Use the Murphy and Topel (1985) two-step correction, or bootstrap both stages jointly

Why not GMM with instruments?

A more natural-looking alternative: simulate $\psi(\varepsilon^{(s)})$, interact with instruments Z , and run classic GMM

$$g_i(\theta) = Z_i \cdot \frac{1}{S} \sum_{s=1}^S \psi(\varepsilon_i^{(s)}; \theta), \quad \hat{\theta} = \arg \min_{\theta} \left(\frac{1}{N} \sum_i g_i(\theta) \right)' W \left(\frac{1}{N} \sum_i g_i(\theta) \right)$$

GMM consistency needs exactly one thing: $E[g_i(\theta_0)] = 0$. Instruments and the weighting matrix W only combine *valid* moments efficiently — they cannot manufacture validity that isn't there.

The moment isn't valid

As $S \rightarrow \infty$, the inner average converges (plain LLN — nonlinearity doesn't break this part) to

$$\mu_i(\theta) \equiv \int \psi(\varepsilon; \theta) f_\varepsilon(\varepsilon) d\varepsilon$$

- ▶ The true orthogonality condition is $E[Z_i \psi_{it}] = 0$, with $\psi_{it} = \psi(\varepsilon_{it}; \theta_0)$ evaluated at firm i , t 's **actual** ε_{it}
- ▶ $\mu_i(\theta_0)$ instead averages $\psi(\varepsilon; \theta_0)$ over *every* candidate ε , weighted only by the marginal f_ε — it never singles out the firm's true ε_{it}
- ▶ Because $\psi(\cdot)$ is nonlinear, $\mu_i(\theta_0) \neq \psi_{it}$ pointwise, and nothing guarantees $E[Z_i \mu_i(\theta_0)] = 0$
- ▶ Missing exactly the reweighting that makes MSL work — $f_\psi(\psi(\varepsilon)) J_i(\varepsilon) f_\omega(\omega(\varepsilon))$, telling you *which* simulated draws are actually plausible for that firm
- ▶ GMM's efficiency machinery operates one level removed from this — it cannot fix an invalid moment, however it's wrapped

Recasting in M doesn't help either

The same trap reappears if we instead simulate $M_{it}^{(s)} = \beta P_t Y_{it} / \rho_t \cdot \mathbf{e}^{-\varepsilon^{(s)}}$ and form $e_{it}^{(s)} = M_{it}^* - M_{it}^{(s)}$ — this looks like it avoids $\mathcal{V}_{it}$, but

$$e(\varepsilon) = M_{it}^* (1 - \mathbf{e}^{-\mathcal{V}_{it} - \varepsilon}) = M_{it}^* - \underbrace{\frac{\beta P_t Y_{it}}{\rho_t} \mathbf{e}^{-\varepsilon}}_{= M(\varepsilon)} = M_{it}^* - M(\varepsilon)$$

using $\mathbf{e}^{-\mathcal{V}_{it}} = \beta P_t Y_{it} / (\rho_t M_{it}^*)$. **It's the identical map** — $\mathcal{V}_{it}$'s information is just M_{it}^* and $P_t Y_{it} / \rho_t$, so the M -version reconstructs it implicitly. Same nonlinearity, same missing reweighting, same problem.

What a valid simulated-moments design would need

- ▶ Match a **data-computable, ψ -free** population object against a forward-simulated analog — e.g. $E[\mathcal{V}_{it}|Z] = E[u_{it}|Z]$ is directly computable from data (since $\varepsilon \perp Z$), while $u^{(s)} = u(e^*(\psi^{(s)}, \omega^{(s)}; \theta))$ can be simulated forward from trial $(\psi^{(s)}, \omega^{(s)})$ draws
- ▶ This ties back to the deconvolved f_u from stage 1 as the natural target
- ▶ Delivers θ but not f_ψ , and needs its own machinery — no real gain over quadrature-MLE, which is already a cheap 1-D integral
- ▶ **Not pursued further** — MSL/quadrature is the headline strategy

Implicit Assumptions

- ▶ The functional forms for $q(e)$ and $\kappa(e)$ are correctly specified and do not change over time
- ▶ No entry, no exit
- ▶ Corporations never evade
- ▶ Partial equilibrium, Inelastic demand, (No GE effects)
 - ▶ Tax increases do not change optimal input choices and output. FOC of materials does not depend on tax rate
 - ▶ Firms are already assumed to be price takers

What about τ ?

- ▶ The data includes sales taxes paid on purchases, $\tau \rho_t M_{it}^*$ and sales taxes paid on sales, $\tau P_t Y_{it}$.
- ▶ I can estimate at the firm level the sales tax rate on sales $\tau_{S,it} = \frac{\tau P_t Y_{it}}{P_t Y_{it}}$ and the sales tax rate on purchases $\tau_{P,it} = \frac{\tau \rho_t M_{it}^*}{\rho_t M_{it}^*}$.
- ▶ I can also estimate the reported effective sales tax rate $\tau_{R,it} = \frac{\tau P_t Y_{it} - \tau \rho_t M_{it}^*}{P_t Y_{it}} = \tau_{S,it} - \tau_{P,it} \frac{\rho_t M_{it}^*}{P_t Y_{it}}$
- ▶ and the *corrected* effective sales tax rate $\tau_{C,it} = \tau_{S,it} - \tau_{P,it} \frac{\rho_t M_{it}^*}{P_t Y_{it}} = \tau_{S,it} - \tau_{P,it} \beta e^{-\varepsilon_{it}}$
 - ▶ I use in regressions $\tau_{\tilde{C},it} = \tau_{S,it} - \tau_{P,it} \beta$

Is the variation in the different estimated tax rates relevant

- ▶ I use the following regression model

$$\mathcal{V}_{it} = \gamma_j + \gamma_t + \tau_{X,it} + \zeta_{it}$$

- ▶ where γ_j are industry fixed effects,
- ▶ γ_t are year fixed effects, and
- ▶ $\tau_{X,it}$ is one of the estimated tax rates ($\tau_{S,it}$, $\tau_{P,it}$, $\tau_{R,it}$, or $\tau_{\tilde{C},it}$).

Variation in estimated τ predicts variation in tax evasion across firms

Dependent Variable: Model:	(1)	(2)	$\mathcal{V}$ (3)	(4)	(5)
<i>Variables</i>					
τ_S	-0.2804* (0.1333)		0.5053*** (0.1422)		
τ_P		-1.890*** (0.1252)	-2.026*** (0.1258)		
τ_R				-1.387* (0.6846)	
τ_C					0.9046*** (0.1803)
<i>Fixed-effects</i>					
Industry (3-digit)	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>					
Observations	40,490	40,505	40,490	40,490	40,490
R ²	0.04767	0.09016	0.09215	0.07638	0.05335
Within R ²	0.00068	0.04532	0.04736	0.03081	0.00665

Clustered (plant & Year) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Couterfactual Question

What Would Be The Optimal Tax Rate?

The government faces a trade-off:

- ▶ as the government increases the tax rate, the tax revenue increases
- ▶ as the tax rate increases, the marginal benefit of evading taxes increase leading to higher levels of tax evasion which reduces tax revenue

What is the optimal rate that maximizes expected tax revenue for the government?

What is the tax rate that lead to median (media) tax evasion of zero?

The Government's Problem

The government wants to choose the sales tax rate τ that maximizes the expected tax revenue from the sum of all firm i

$$\max_{\tau} \sum_i q(e) \tau [P_t Y_{ijt} - \rho_t M_{ijt}] + (1 - q(e)) \tau [P_t Y_{ijt} - \rho_t (M_{ijt} + e_{ijt})]$$

Simplifying,

$$\max_{\tau} \sum_i \tau [P_t Y_{ijt} - \rho_t M_{ijt} - (1 - q(e)) \rho_t e_{ijt}]$$

where the first two terms are the potential tax revenue (in the absence of tax evasion) and the last terms is the expected undetected tax evasion.

for firm i ,

$$\begin{aligned} & [P_t Y_{ijt} - \rho_t M_{ijt} - (1 - q(e)) \rho_t e_{ijt}] \\ & + \tau \left[-(1 - q(e)) \rho_t \frac{\partial e_{ijt}}{\partial \tau} + \rho_t e_{ijt} \frac{\partial q(e)}{\partial e} \frac{\partial e_{ijt}}{\partial \tau} \right] = 0 \end{aligned}$$

simplifying,

$$\begin{aligned} P_t Y_{ijt} - \rho_t M_{ijt} &= (1 - q(e)) \rho_t e_{ijt} \\ &+ \tau \rho_t \frac{\partial e_{ijt}}{\partial \tau} \left[(1 - q(e)) - \frac{\partial q(e)}{\partial e} e_{ijt} \right] \end{aligned}$$

Numerical Strategy

1. Stage 2 is already done by this point: $\theta = (\lambda, \delta_0, \delta_1, \delta_2)$ and f_ψ are estimated (MSL/MSM) and held fixed for everything below — no new estimation happens in the counterfactual
2. For each firm-year, recover the implied **type** (ω, ψ) at the *observed* τ , using the same map as the likelihood (Equation 19): for $M_{it}(\varepsilon) = M_{it}^* \mathbf{e}^{-\mathcal{V}_{it}-\varepsilon}$,

$$\omega(\varepsilon) = \tilde{\mathcal{W}}_{it} - (1 - \beta)\varepsilon, \quad \psi(\varepsilon) = h(e(\varepsilon)) - \delta_0 + \delta_1\omega(\varepsilon) - \delta_2\omega(\varepsilon)^2$$

3. Hold (ω, ψ) fixed as firm primitives — they don't respond to the tax rate — and resolve the evasion FOC at a proposed $\tilde{\tau}$. Under linear q , closed form, no root-finding:

$$e'(\tilde{\tau}; \omega, \psi) = \frac{\tilde{\tau}\rho_t - C(\omega, \psi)}{2\lambda\tilde{\tau}\rho_t}, \quad C(\omega, \psi) \equiv \mathbf{e}^{\delta_0 - \delta_1\omega + \delta_2\omega^2 + \psi}$$

4. e' is a **deterministic function of the same** ε that generated (ω, ψ) — not an independent latent. So expected revenue is a *single* integral over ε , weighted exactly as the likelihood weights it, normalized by the firm's own likelihood contribution $\mathcal{L}_i(\theta)$ (already computed in stage 2) so this is a genuine conditional expectation given firm i 's data:

$$E[R_i(\tilde{\tau})] = \frac{1}{\mathcal{L}_i(\theta)} \int \tilde{\tau} \left[P_t Y_{it} - \rho_t M_{it}(\varepsilon) - (1 - q(e'(\varepsilon))) \rho_t e'(\varepsilon) \right] f_\psi(\psi(\varepsilon)) J_i(\varepsilon) d\varepsilon \quad (22)$$

5. Evaluate Equation 22 with the same quadrature nodes used in stage 2 — reuse the machinery, don't refit anything
6. Find $\tilde{\tau}^* = \arg \max_{\tilde{\tau}} \sum_i E[R_i(\tilde{\tau})]$ by grid or golden-section search over the scalar $\tilde{\tau}$ (plot the objective first to confirm it's single-peaked before reporting a point estimate)

Alternative: Simulated Draws (MSM route)

If stage 2 used MSM instead of MSL, f_ψ is never recovered — only θ . The quadrature weight in Equation 22 is unavailable, but $(\omega(\varepsilon), \psi(\varepsilon))$ and $e'(\tilde{\tau}; \omega, \psi)$ only ever need θ , so reuse the same simulated draws MSM already used for moment-matching:

$$E[\widehat{R_i}(\tilde{\tau})] = \frac{1}{S} \sum_{s=1}^S \tilde{\tau} \left[P_t Y_{it} - \rho_t M_{it}(\varepsilon^{(s)}) - (1 - q(e'(\varepsilon^{(s)}))) \rho_t e'(\varepsilon^{(s)}) \right], \quad \varepsilon^{(s)} \sim$$

- ▶ No f_ψ anywhere — consistent with what MSM delivers
- ▶ Every ε draw is treated as equally likely (weighted only by the unconditional f_ε); the quadrature version additionally down-weights ε that would imply an implausible cost shock given f_ψ — same MSL-vs-MSM efficiency trade-off, now downstream in the counterfactual
- ▶ Use **common random numbers**: draw $\varepsilon^{(s)}$ once per firm and reuse the same draws across every candidate $\tilde{\tau}$, or the search over $\tilde{\tau}$ chases simulation noise instead of the Laffer curve

Estimating Tax Evasion by Deconvolution

Objective

- ▶ Empirical exercise to motivate structural model

What can be deconvolved and what can't

Convolution (Meister, 2009)

The density of the sum of two **independent** random variables is equal to the **convolution** of the densities of both addends; hence

$$f_V = \int f_\varepsilon(V - u)f_u(u)du$$

Under $\varepsilon \perp\!\!\!\perp M$, e and $e \perp\!\!\!\perp M$, with known f_ε and $u = \ln\left(\frac{M+e}{M}\right)$. Also $M^* = M + e$, and $V = u + \varepsilon$.

Then, I can deconvolve:

► f_u : $f_V = \int f_\varepsilon(V - u)f_u(u)du$

► $f_{\ln M}$: By letting $V - \ln M^* = -\ln M + \varepsilon$, I can deconvolve:
 $f_{V - \ln M^*} = \int f_\varepsilon(V - \ln M^* + \ln M)f_{\ln M}(\ln M)d \ln M$

Deconvolving the overreporting log-ratio

- ▶ Deconvolution recovers the distribution of the **overreporting log-ratio**

$$u_{it} \equiv \ln \left(1 + \frac{e_{it}}{M_{it}} \right), f_u.$$

- ▶ the *share of overreported over true inputs* in logs
- ▶ Same object discussed previously, no change in estimation
- ▶ Clear link to model:
 - ▶ the structural e is explicitly identified inside the empirically recovered object,
 - ▶ rather than u standing in as an alternative definition of tax evasion

$$\mathcal{V}_{it} = u_{it} - \varepsilon_{it} \tag{23}$$

Plotting $\mathcal{V}$ and $\varepsilon \hookrightarrow$

Going beyond

We said $M^* \not\perp e, M$, but we have more than that:

► There is *deterministic mapping* that is **monotonic** and **differentiable** (C^1)

Theorem 1.7.1 (Hogg et al. 2019): Let X be a continuous random variable with pdf $f_X(x)$ and support $\mathcal{S}_X$. Let $Y = g(X)$ where $g(x)$ is a one-to-one [*monotonic*] *differentiable* function on the support of X , $\mathcal{S}_X$. Denote the inverse of g by $x = g^{-1}(y)$ and $dx/dy = d[g^{-1}(y)]/dy$. Then, the pdf of Y is given by

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|, \text{ for } y \in \mathcal{S}_Y$$

where the support of Y is the set $\mathcal{S}_Y = \{y = g(x) : x \in \mathcal{S}_X\}$

Getting $f_{\frac{e}{M}}$ and f_e

Therefore, with the f_u density on hand, we can get $f_{\frac{e}{M}}$ or f_e by a **density transformation**, using the identities

$$u = \ln \left(1 + \frac{e}{M} \right) \quad u = \ln \left(\frac{M^*}{M^* - e} \right)$$

With $X = u$, $Y = \frac{e}{M}$, and $g^{-1}(y) = \ln(1 + y)$

$$f_{\frac{e}{M}}(y) = f_u(\ln(1 + y)) \left| \frac{1}{1 + y} \right| \quad (24)$$

and with $Y = e$, and $g^{-1}(y) = \ln \left(\frac{M+y}{M} \right)$

$$f_e(y) = f_u \left(\ln \left(\frac{M^*}{M^* - y} \right) \right) \left| \frac{1}{M^*} \right| \quad (25)$$

Informal proof

If $U = g(R)$ for a strictly monotonic differentiable g , then

$$f_R(r) = f_U(g(r))|g'(r)|$$

Proof (via CDFs). Take g strictly increasing (your case). Then

$$F_R(r) = P(R \leq r) = P(g(R) \leq g(r)) = P(U \leq g(r)) = F_U(g(r)),$$

using that g increasing preserves the inequality direction.

Differentiate both sides in r :

$$f_R(r) = \frac{d}{dr}F_U(g(r)) = f_U(g(r))g'(r).$$

If g were decreasing instead, the inequality flips ($P(U \geq g(r)) = 1 - F_U(g(r))$), and differentiating introduces a minus sign that exactly cancels the sign of $g'(r)$, hence $|g'(r)|$ of the general formula.

Everything goes through unchanged conditional on $Z = z$ (just work with $F_{R|Z}(\cdot | z)$, $F_{u|Z}(\cdot | z)$ throughout).

Deconvolution estimates the unconditional f_u because of how we set it up

- ▶ From f_u , we can learn the probability of a firm of overreporting u ,
 $P(U < u) = \int_0^u f_u(t)dt$, or
 - ▶ the expected overreporting $\mu_u = \int u f_u(u)du$
- ▶ This is the unconditional probability, the same for all firms regardless of their characteristics
- ▶ We might be interested in the probability of overreporting conditional on the firm size

What if we are interested in the conditional density?

$$\begin{aligned} f_{V|Z}(V | Z) &= \frac{f_{V,Z}(V, Z)}{f_Z(Z)} \\ &= \int f_{V|u,Z}(V | u, Z) f_{u|Z}(u | Z) du \\ &= \int f_{\varepsilon}(V - u) f_{u|Z}(u | Z) du \end{aligned}$$

using

- ▶ $f_{V|u,Z}(V | u, Z) = f_{\varepsilon|u,Z}(V - u | u, Z) - V = u + \varepsilon$ conditional on u
- ▶ $\varepsilon \perp (u, Z)$ **jointly**, so $f_{\varepsilon|u,Z} = f_{\varepsilon}$

with $u \not\perp Z$ (otherwise conditioning on Z would be pointless).

Estimating the Conditional distribution

- ▶ **Discrete/bucketed** Z (e.g. quintiles of K_{it} or L_{it}): the same 1-D deconvolution separately within each bucket.
 - ▶ We've already looked into $E[u|Z]$
- ▶ **Continuous** Z : nonparametric deconvolution with a continuous covariate
 - ▶ Literature: $f_{u|V}(u|V)$ (Wang and Ye 2015), $E[Z|u]$ (Fan and Truong 1993), and $f_{Z|u}$ (Huang and Zhou 2020)
 - ▶ $f_{u|Z}$: feasible using characteristic functions and kernels
- ▶ The previous argument to recover $f_{e|Z}$ and $f_{e/M|Z}$ from $f_{u|Z}$ follows by the previous argument on the unconditional density
- ▶ Ideally, what I want is $f_{e|\omega}$, but that's what f_{ψ} is, the object I will obtain with the structural approach

Conditional Deconvolution — Backup

What About the Full Distribution?

- ▶ So far: $E[u_{ijt} \mid Z_{ijt}]$, and only at 10 decile bins of Z
- ▶ Natural next question: the whole conditional density $f_{u|Z}(u \mid z)$, for Z continuous
 - ▶ Is the size gradient in the *level* of evasion, its *spread*, or its *shape*?

Aside: How Deconvolution Works

We never observe u_{ijt} directly — only $V_{ijt} = u_{ijt} + \varepsilon_{ijt}$, a “blurred” version of it

- ▶ A histogram of V mixes the true shape of u with the noise ε — the density of a sum of independent things is the *convolution* of their densities, $f_V = f_u * f_\varepsilon$
- ▶ Convolutions are hard to undo directly, but they become simple *multiplication* after a Fourier transform (the characteristic function):

$$\phi_V(t) = \phi_u(t) \phi_\varepsilon(t) \quad \implies \quad \phi_u(t) = \frac{\phi_V(t)}{\phi_\varepsilon(t)}$$

- ▶ Estimate ϕ_V from the data, divide by the already-known ϕ_ε , transform back $\rightarrow \hat{f}_u$
- ▶ This is exactly the (unconditional) trick behind the first-stage results (Carroll and Hall 1988; Stefanski and Carroll 1990)

Conditional Version: Add a Local Window

Same trick, done separately for each value of Z :

- ▶ Instead of using *all* firms to build one blurred distribution, use only firms with Z_i close to z — a moving local window, weighted by a kernel K
- ▶ Un-blur *that* local distribution the same way as before

$$\hat{\phi}_{V|Z=z}(t) = \frac{\sum_i K\left(\frac{Z_i - z}{h}\right) e^{itV_i}}{\sum_i K\left(\frac{Z_i - z}{h}\right)} \implies \hat{f}_{u|Z=z}(u) = \frac{1}{2\pi} \int e^{-itu} \frac{\hat{\phi}_{V|Z=z}(t)}{\phi_\varepsilon(t)} \omega(t)$$

- ▶ $\omega(t)$: a damping weight, standard in any deconvolution, to keep the division numerically stable
- ▶ Valid because $\varepsilon \perp (u, Z)$: within any Z -window, the noise is still independent of u — the same assumption used throughout this section

Is This Standard?

- ▶ (Fan and Truong 1993), already cited on the “is it feasible?” slide, addresses the *mirror-image* problem: a noisy covariate and a clean outcome (see also (Huang and Zhou 2020) for the conditional-density version of that same direction)
- ▶ Ours runs the other way — clean Z , noisy u — which is why the conditional *mean* never needed correcting: $\varepsilon \perp Z \Rightarrow E[V \mid Z] = E[u \mid Z]$
- ▶ Only the *density* needs deconvolving, and the tool is just the classical deconvolution kernel (Carroll and Hall 1988; Stefanski and Carroll 1990), applied inside a local window — a natural combination, not (to our knowledge) a named result in the literature

Do We Need the Full Density?

- ▶ The conditional mean already carries the headline story: evasion rises then falls with size — going further only pays off if it answers a question the mean can't
- ▶ The one moment that would matter: the conditional **variance** $\text{Var}(u \mid Z = z)$
 - ▶ Does “large firms don't evade” mean (a) *every* large firm individually evades a little — tight distribution near zero, like the corporations — or (b) a mix of some firms still evading heavily and others near zero that cancel out in the mean?
 - ▶ (a) vs. (b) is a real audit-targeting question: under (a), size alone is a good targeting rule at the top; under (b) it isn't — some large firms are still hiding substantial evasion behind the average
- ▶ If pushed further: the full nonparametric conditional deconvolution (previous slides) is the general method; cumulant/method-of-moments matching is a cheap over-identification check on the variance estimate, not a substitute for it

The Geography of Tax Evasion

How Does a Firm's Location Affect its Tax Evasion?

Firms located in different regions might face different probabilities of detection and different tax rates, and thus different incentives to evade taxes.

- ▶ Tax rates:
 - ▶ Departments and municipalities in Colombia can set their own tax rates similar to sales tax, for example the Industry and Commerce Tax
 - ▶ Special tax regimes for firms located in certain regions, such as the Free Trade Zones (Zonas Francas) and the Special Economic Zones (Zonas Económicas Especiales)
 - ▶ Special ownership model in Atlantic coast
 - ▶ Some variation in tax rates is observed in the data

How Does a Firm's Location Affect its Tax Evasion?

- ▶ Probability of detection: Two hypothesis
 1. Firms located near the tax authority's offices might have access to the tax authority (lobbying, bribery, etc.) and thus face a lower probability of detection
 2. Firms located farther away from the tax authority's offices might face a lower probability of detection, because the auditing cost for the tax authority increases with distance, and thus they are less likely to be audited

Data

In the dataset from Colombia, I have two geographic variables:

- ▶ `section_country_code`: Section of the country (eg. Antioquia, Atlantico, Bogota D.E., etc.)
- ▶ `metro_area_code`: Metropolitan Area (eg. Bogota, Cali, Medellin, Barranquilla).

Metro Area Summary Statistics

Metro Area	N	Market Share	Avg. Sales Tax Rate	Med. Sales Tax Rate	Avg. Tax Evasion	Med. Tax Evasion
Bogotá D.C., Soacha	14345	35.3	7.4	8.9	4.2	10.3
Medellín, Valle de Aburrá	8147	19.9	7.6	9.2	4.6	10.7
Rest of Colombia	5895	14.9	4.6	2.0	0.7	10.2
Cali, Yumbo	4538	11.4	7.2	8.4	4.2	10.1
Barranquilla, Soledad	2896	7.3	5.6	5.9	2.5	13.3
Bucaramanga, Giron, Floridablanca	2311	5.3	5.7	6.0	5.9	13.8
Pereira, Santa Rosa de Cabal, Dosquebradas	1128	2.7	7.6	9.1	11.1	21.1
Manizales, Villamaria	639	1.6	7.0	8.1	3.2	7.2
Cartagena	606	1.6	4.6	2.7	11.6	20.0

Section of the Country Summary Statistics

Depto	N	Market Share	Avg. Sales Tax Rate	Med. Sales Tax Rate	Avg. Tax Evasion	Med. Tax Evasion
Bogotá D.C.	13954	34.3	7.3	8.9	4.3	10.3
Antioquia	8467	20.7	7.5	9.1	3.4	9.7
Valle del Cauca	5780	14.6	6.7	7.0	4.1	10.3
Atlántico	2922	7.3	5.6	5.8	2.6	13.2
Santander	2585	6.0	5.6	6.0	5.5	12.8
Cundinamarca	1282	3.5	5.8	6.0	1.9	12.8
Risaralda	1161	2.8	7.4	8.9	11.0	21.0
Tolima	763	1.9	4.0	0.0	10.1	21.2
Caldas	710	1.8	6.7	6.8	-0.8	6.6
Bolívar	635	1.7	4.4	2.1	9.8	19.7
Norte de Santander	617	1.4	5.2	6.0	-7.6	2.9
Boyacá	290	0.7	5.8	6.0	1.8	8.8
Quindío	288	0.7	5.2	6.0	6.9	18.1
Nariño	255	0.6	3.4	0.0	14.7	25.1
Huila	182	0.5	4.4	0.0	10.1	19.2
Cauca	155	0.4	3.7	0.0	-0.4	-1.8
Meta	119	0.3	2.8	0.0	-10.5	12.4
Córdoba	126	0.3	3.7	0.3	21.1	25.1
Cesar	72	0.2	1.8	0.0	-11.8	16.9
Magdalena	73	0.2	5.0	1.3	-33.7	-13.2
Sucre	54	0.1	4.5	4.4	-5.7	-11.4
Putumayo	9	0.0	8.4	6.0	22.8	7.1
Caquetá	5	0.0	5.9	6.0	50.7	70.1
Chocó	1	0.0	3.1	3.1	-36.0	-36.0

Prelim Results (Metro Area)

Dependent Variable:	$\mathcal{V}$
Model:	(1)
<i>Variables</i>	
Bogotá D.C., Soacha	0.0485*** (0.0132)
Cali, Yumbo	0.0706*** (0.0174)
Medellín, Valle de Aburrá	0.0440** (0.0141)
Manizales, Villamaria	0.0359 (0.0512)
Barranquilla, Soledad	0.0309 (0.0266)
Bucaramanga, Giron, Floridablanca	0.0654** (0.0226)
Pereira, Santa Rosa de Cabal, Dosquebradas	0.1391*** (0.0355)
Cartagena	0.1402* (0.0758)
Rest of Colombia	0.0244 (0.0197)
<i>Fit statistics</i>	
Observations	32,373
R ²	0.00280
Adjusted R ²	0.00255

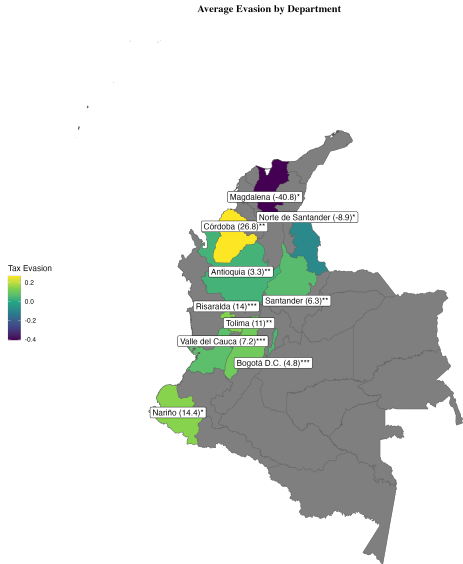
Clustered (plant & year) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Prelim Results (Departamento)

Dependent Variable: Model:	ψ (1)
<i>Variables</i>	
Antioquia	0.0331** (0.0143)
Atlántico	0.0323 (0.0267)
Bogotá D.C.	0.0484*** (0.0133)
Bolívar	0.1116 (0.0743)
Boyacá	0.0498 (0.0822)
Caldas	-0.0275 (0.0574)
Cauca	0.1291 (0.0779)
Cesar	-0.3661 (0.3888)
Córdoba	0.2684** (0.0882)
Cundinamarca	0.0371 (0.0449)
Huila	0.1464 (0.0816)
Magdalena	-0.4084* (0.2023)
Meta	-0.1142 (0.1742)
Nariño	0.1441* (0.0655)
Norte de Santander	-0.0894* (0.0427)
Quindío	0.0801 (0.0608)
Risaralda	0.1396*** (0.0349)
Santander	0.0629** (0.0227)
Sucre	0.0642 (0.1590)
Tolima	0.1397** (0.0414)
Valle del Cauca	0.0718*** (0.0152)
<i>Fit statistics</i>	
Observations	32,373
R ²	0.00895
Adjusted R ²	0.00834

Clustered (plant & year) standard errors in parentheses
 Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

On the Map



Next Steps

- ▶ Are there any patterns? Do they make sense?
- ▶ Can any of these regions be linked to Free Trade Zones or Special Economic Zones? Since when are these in place?
 - ▶ Zonas Francas: started in 1958. During this period, there were changes made in 1981 and 1985, and 1991. 1981-1984, firms in Zonas Francas were exempt from income and sales taxes, as well as from import duties. They were under strict vigilance by the tax authority and confined to specific locations. Since to be eligible, your products had to be exported, most traditional firms choose not to locate in ZFs. Mostly multinationals and exporting Maquilas took advantage of ZFs. There were ZFs in Barranquilla, Cartagena, Santa Marta, and Buenaventura.
 - ▶ Zonas Económicas Especiales: started in 2020

Appendix

Appendix

B) Value of Reported Ghost Purchases Over Firm Total Purchases, by Firm Size

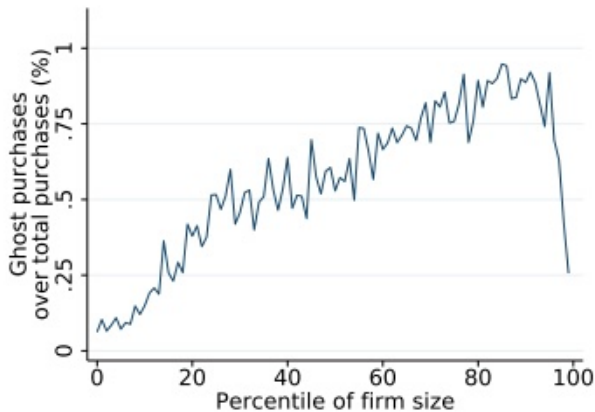


Figure 1: Figure 1B from Carrillo et al. (2022) ↩

Robust Check for Size Heterogeneity

$$\mathcal{V}_{ijt} = \sum_{z \in Z} \sum_{h \in 1,2} \mu_{z,h} x_{z,ijt}^h + \eta_{ijt} \quad (26)$$

Table 1: Coefficient estimates . 95% Bootstrap CI with 250 replications. Stars denote Bootstrap p-value significance at (***) 0.001, (**) 0.05, and (*) 0.1 level.

	IV: Lagged Revenue	OLS: Lagged Revenue
Revenue	1.8012*** [1.5744, 2.0744]	
Revenue ²	-0.7594*** [-1.0265, -0.5149]	
Revenue _{t-1}		1.456*** [1.274, 1.6735]
Revenue _{t-1} ²		-0.5586*** [-0.784, -0.339]
Labour	-0.7149*** [-0.9244, -0.5631]	-0.4281*** [-0.6214, -0.2994]
Labour ²	0.1051 [-0.0561, 0.3035]	-0.09 [-0.2306, 0.1051]
Capital	-0.1124 [-0.277, 0.085]	-0.022 [-0.1776, 0.1737]
Capital ²	-0.2332** [-0.4548, -0.0406]	-0.2997*** [-0.5255, -0.1106]
Exports	-0.1127 [-0.7196, 0.3843]	-0.2759 [-0.8917, 0.192]
Exports ²	0.066 [-0.483, 0.7506]	0.2503 [-0.2569, 0.9386]
Imports	-1.1048*** [-1.4765, -0.7346]	-1.1253*** [-1.5047, -0.7651]
Imports ²	1.3084*** [0.8792, 1.7581]	1.3469*** [0.9235, 1.8055]
Const.	-0.1309*** [-0.1662, -0.0985]	-0.1158*** [-0.1492, -0.0826]

Table 2: Coefficient estimates. Percentile of size of different variables. 95% Bootstrap CI with 250 replications. Stars denote Bootstrap p-value significance at (***) 0.001, (**) 0.05, and (*) 0.1 level.

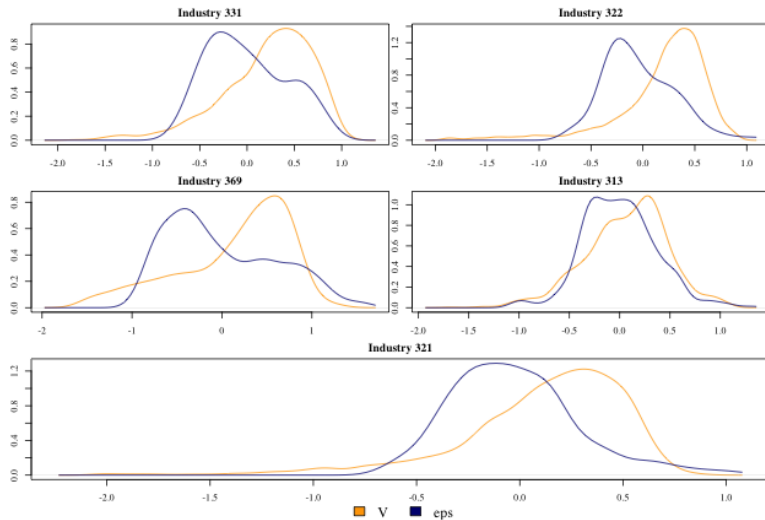
	IV: Lagged Revenue	OLS: Lagged Revenue
Revenue	1.1491*** [1.0863, 1.2497]	
Revenue _{t-1}		0.9948*** [0.9362, 1.0738]
Labour	-0.6258*** [-0.7001, -0.5599]	-0.5044*** [-0.5691, -0.4402]
Capital	-0.3082*** [-0.3679, -0.2278]	-0.2737*** [-0.3309, -0.1943]
Exports	-0.1004 [-0.5756, 0.283]	-0.2292 [-0.7139, 0.1385]
Imports	-0.7292*** [-0.9883, -0.5182]	-0.7386*** [-1.0034, -0.5334]

Why the Negative Sign for Some Small Firms?

- ▶ Smaller firms are more likely to be Proprietorships and less formal
- ▶ Proprietorships report their industrial activity as part of their individual income taxes. They need to keep track of their personal expenses and their industrial activity expenses
- ▶ Because they are more likely to be less formal than larger firms, they may lack a rigorous expense report system
- ▶ Thus, they might be consistently underreporting their expenses

Densities of $\mathcal{V}$ and ε

V and Epsilon Densities by Industry



Empirical Exercises using $\mathcal{V}$

Empirical exercises remain valid as long as changes in $\mathcal{V}$ (tax evasion proxy) are not explained by changes in M (or ε) but only by e or e/M

- ▶ Changes in the proxy are explained by changes to incentives to adjust overreporting not firms adjusting their optimal choice of true inputs.
- ▶ Fiscal Reform of 1983:
 - ▶ Alternative argument, changes in true inputs explain the increase in the overreporting share. How? Suppose e is unchanged. This implies that after the policy reform unincorporated firms decrease their level of inputs by either an increase in relative prices or productivity shocks. Productivity shocks are unrelated to tax rates, to this is unlikely. According to the FOC, taxes do not show in the optimal choice of firms. Besides, we see an increase after the reform, while the reform made only a one-time change. So this alternative argument does not hold.
 - ▶ Our argument is more transparent: an increase in the sales tax rate lead to an increase in the overreporting share because a higher tax rate increases the marginal benefit of overreporting.

Empirical Exercises using $\mathcal{V}$ (Continued)

- ▶ Tax Evasion heterogeneity along firm size:
 - ▶ Alternative argument, correlation between true inputs and size explain why the overreporting share increases along the size distribution and drops at the top. How? fix e . CD could not be capturing the relationship between labour (or capital) and materials. A non-linear relationship could explain the pattern in the data.
 - ▶ However, the pattern is consistent with evidence from other papers with administrative data. As a robustness check, I also use the lag and double lag labour and capital and the pattern persist.

References

- Allingham, Michael G., and Agnar Sandmo. 1972. "Income Tax Evasion: A Theoretical Analysis." *Journal of Public Economics* 1 (November): 323–38. [https://doi.org/10.1016/0047-2727\(72\)90010-2](https://doi.org/10.1016/0047-2727(72)90010-2).
- Carrillo, Paul, Dave Donaldson, Dina Pomeranz, and Monica Singhal. 2022. *Ghosting the Tax Authority: Fake Firms and Tax Fraud*. NBER. <http://www.nber.org/papers/w30242>.
- Carroll, Raymond J., and Peter Hall. 1988. "Optimal Rates of Convergence for Deconvolving a Density." *Journal of the American Statistical Association* 83 (404): 1184–86. <https://doi.org/10.1080/01621459.1988.10478718>.
- Fan, Jianqing, and Young K. Truong. 1993. "Nonparametric Regression with Errors in Variables." *The Annals of Statistics* 21 (4): 1900–1925. <https://doi.org/10.1214/aos/1176349402>.